



04 — Graph Manager (Prototype Edges)

Overview

The graph manager maintains a **sparse directed graph** over semantic prototypes. Each edge encodes a weighted relationship between two prototypes derived from three signals: semantic similarity, temporal transition probability, and co-occurrence frequency.

Data Structure

A directed edge ($p \rightarrow q$) with components:

$$\text{edge}(p \rightarrow q) = (w_{\text{sim}}, w_{\text{trans}}, w_{\text{coact}}, w, \text{ts})$$

Where $w = \lambda_1 w_{\text{sim}} + \lambda_2 w_{\text{trans}} + \lambda_3 w_{\text{coact}}$ is the fused weight used for retrieval.

Mathematical Formulation

Edge Weight Fusion

$$w_{pq} = \lambda_1 \cdot \cos(\mathbf{c}_p, \mathbf{c}_q) + \lambda_2 \cdot P(q | p) + \lambda_3 \cdot \text{coact}(p, q)$$

Where:

- $\cos(\mathbf{c}_p, \mathbf{c}_q)$ = cosine similarity of prototype centroids
- $P(q | p)$ = empirical transition probability (EMA-accumulated)
- $\text{coact}(p, q)$ = co-occurrence count (EMA-accumulated)

EMA Accumulation

For transition and coactivation components, each replay pass produces a raw count; the stored value is an EMA:

$$v_{\text{new}} = v_{\text{old}} \cdot \alpha_d + \text{count}_{\text{current}} \cdot (1 - \alpha_d)$$

Where $\alpha_d = \text{accumulate_decay}$.

Interpretation: `accumulate_decay = 0.3` means 70% of the stored weight comes from the most recent pass, 30% from prior history. This makes the graph quickly responsive to new patterns while retaining some stability.

Homeostatic Normalization

Per-source prototype, outgoing edges are normalized to a target L1 sum:

$$w_{pq} \leftarrow w_{pq} \cdot \frac{T}{\sum_{q'} w_{pq'}}, \quad T = \text{homeostatic_target}$$

This prevents super-hubs (frequently co-occurring prototypes) from dominating all outgoing activation.

Relative Pruning

After normalization, edges are pruned if:

$$w_{pq} < \max(r_{\text{ratio}} \cdot \max_{q'} w_{pq'}, \tau_{\text{abs}})$$

Where:

- $r_{\text{ratio}} = \text{prune_ratio}$ — relative to strongest outgoing edge
- $\tau_{\text{abs}} = \text{prune_below}$ — absolute floor

Neighbor Retrieval

For spreading activation, the top-k outgoing edges from a prototype are fetched:

$$\mathcal{N}(p, k) = \{(q, w_{pq}) \mid q \in \text{top-}k \text{ by } w_{pq} \text{ DESC}\}$$

Where $k = \text{neighbor_top_k}$ (used in activation propagation, default: `6`).

Configuration

GraphConfig

Parameter	Type	Default	Description
<code>top_k_similarity</code>	<code>int</code>	<code>8</code>	Max similarity edges per prototype
<code>top_k_coactivation</code>	<code>int</code>	<code>6</code>	Max coactivation edges per prototype
<code>prune_below</code>	<code>float</code>	<code>0.05</code>	Absolute pruning threshold
<code>lambda_similarity</code> λ_1	<code>float</code>	<code>1.0</code>	Weight for cosine similarity component
<code>lambda_transition</code> λ_2	<code>float</code>	<code>0.5</code>	Weight for transition probability
<code>lambda_coactivation</code> λ_3	<code>float</code>	<code>0.3</code>	Weight for coactivation component
<code>accumulate_decay</code> α_d	<code>float</code>	<code>0.3</code>	EMA decay (lower = faster adaptation)
<code>homeostatic_enabled</code>	<code>bool</code>	<code>True</code>	Enable homeostatic L1 normalization
<code>homeostatic_target</code> T	<code>float</code>	<code>0.5</code>	Target L1 sum per source
<code>prune_ratio</code> r_{ratio}	<code>float</code>	<code>0.2</code>	Relative pruning fraction of max

Key Invariants

1. Edges are directed — $w_{pq} \neq w_{qp}$ in general.
2. The fused weight w is what's used in retrieval; component weights are stored for diagnostics.
3. Homeostatic normalization runs after every accumulation pass.
4. `accumulate_decay = 0.3` is aggressive — single-exposure traces fade quickly unless reinforced.
5. Similarity edges dominate the weight ($\lambda_1 = 1.0$) while transition ($\lambda_2 = 0.5$) and coactivation ($\lambda_3 = 0.3$) are secondary signals.